

BST Physics Curriculum Vol 13 Ch 6 — Biochemistry Bridge (Vol 12 Cross-Link) v0.4 (Calibration #23 substance-floor refill)

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redirect)

Vol 13 Chapter 6 — Biochemistry Bridge (Vol 12 Cross-Link)

Chapter motivation

Biochemistry is chemistry at biological scale. The transition from “chemistry” (Vol 12) to “biology” (Vol 13) is continuous — there is no fundamental separation between the two domains, only the addition of: (1) catalytic enzymes (proteins as highly specific reaction accelerators), (2) organized metabolic pathways (sequenced reactions producing energy + biosynthetic intermediates), (3) ATP energy currency cycle (energy storage + transfer molecule), (4) information-directed synthesis (DNA-RNA-protein central dogma per Vol 13 Ch 2). Standard biochemistry treatment (Berg/Tymoczko/Stryer, Lehninger, Voet): enzyme kinetics + Michaelis-Menten + active sites; metabolic pathways (glycolysis + TCA cycle + oxidative phosphorylation); chemiosmotic theory (Mitchell 1961 Nobel 1978); membrane bioenergetics. BST substrate-cartography reading: biochemistry IS Vol 12 chemistry at biological scale; biochemical reactions are substrate K-type transitions inheriting BST primary integer patterns. The bridge between Vol 12 and Vol 13 is the same substrate D_{IV}^5 at the same K-type representation theory level — biology adds organization without adding new physics.

Section 6.0b — Reader-grade 3-level pedagogy

Level 1: Biochemistry = Vol 12 chemistry + enzymes + metabolic pathways + ATP cycle + information-directed synthesis. Biochemical reactions = substrate K-type transitions same as Vol 12; biology adds organization not new physics. Enzymes = highly specific substrate K-type pathway shortcuts (Vol 12 Ch 5 catalysis full enzymatic case).

Level 2 (graduate-biochemist accessible): Enzymes: protein catalysts with three defining features — (1) catalytic power (rate enhancements 10^6 - $10^{20}\times$ over uncatalyzed reactions, Wolfenden table); (2) substrate specificity (lock-and-key Fischer 1894 or induced-fit Koshland 1958); (3) regulation (allosteric per Monod-Wyman-Changeux 1965, feedback inhibition, covalent modification). Michaelis-Menten kinetics: $v = V_{\text{max}} [S]/(K_M + [S])$ with $V_{\text{max}} = k_{\text{cat}} [E_{\text{total}}]$ and K_M characteristic concentration; Lineweaver-Burk double-reciprocal plot linearizes. Active site = small region of enzyme (~20-50 residues) where substrate binds + reaction occurs; transition-state stabilization (Pauling 1948) is core catalytic mechanism. Metabolic pathways: organized sequences of enzyme-catalyzed reactions producing energy + biosynthetic intermediates. Glycolysis (Embden-Meyerhof-Parnas pathway, 1930s): glucose $\rightarrow$ 2 pyruvate + 2 ATP + 2 NADH; 10 enzyme steps; near-universal across life. TCA cycle (Krebs cycle 1937 Nobel 1953): pyruvate (via acetyl-CoA) $\rightarrow$ CO₂ + NADH + FADH₂ + GTP; 8 enzyme steps; central metabolic hub. Oxidative phosphorylation: NADH + FADH₂ donate electrons to electron transport chain (4 complexes); proton gradient across inner mitochondrial membrane; ATP synthase uses proton motive force to phosphorylate ADP $\rightarrow$ ATP. Chemiosmotic theory (Mitchell 1961, Nobel 1978): membrane proton gradient is the universal energy-coupling mechanism in bioenergetics. ATP cycle: ATP + H₂O $\rightarrow$ ADP + Pi has $\Delta G^\circ \approx -30$ kJ/mol; couples to non-spontaneous biosynthesis + active transport + mechanical work; $\sim 10^8$ ATP molecules per cell per second turnover. BST substrate-cartography reading: enzymes = highly specific substrate K-type pathway shortcuts (Vol 12 Ch 5 catalysis at extreme specificity); metabolic pathways = substrate Zone-2 commitment-cycle sequences; ATP cycle = substrate Zone-2/Zone-3 energy currency carrier. Chemiosmotic proton gradient inherits substrate proton-electron mass ratio $m_p/m_e = 6\pi^5$ (Vol 13 Ch 4 DNA proton siblings). Biochemistry IS Vol 12 chemistry at biological scale — biology adds organization without adding new physics.

Level 3 (5th-grader accessible): Biochemistry is just chemistry happening inside living cells. The same chemical reactions you learned about in chemistry class (Vol 12) happen in your body, with three additions: (1) enzymes are super-fast specific helpers (proteins that speed up reactions by millions or billions of times); (2) metabolic pathways organize reactions into ordered sequences (like assembly lines); (3) ATP is the cell's universal energy currency (made in mitochondria, used everywhere). BST framework says biochemistry uses the same substrate D_IV⁵ as all chemistry — biology adds organization and information, not new physics.

Section 6.1 — Enzymes (Protein Catalysts)

Enzymes catalyze biological reactions with three key features: 1. Catalytic power: rate enhancements 10^6 - $10^{20}\times$ (Wolfenden 2001 table for prominent enzymes) 2. Substrate specificity: lock-and-key (Fischer 1894) or induced-fit (Koshland 1958) 3. Regulation: allosteric (Monod-Wyman-Changeux 1965), feedback inhibition, covalent modification

Active site (~20-50 residues) provides transition-state stabilization (Pauling 1948

core mechanism).

BST framing: enzymes = highly specific substrate K-type pathway shortcuts (Vol 12 Ch 5 catalysis at extreme specificity).

Section 6.2 — Michaelis-Menten Kinetics

$v = V_{\max} [S]/(K_M + [S])$ with $V_{\max} = k_{\text{cat}} [E_{\text{total}}]$ and $K_M = (k_{-1} + k_{\text{cat}})/k_1$ characteristic concentration.

At $[S] \gg K_M$: $v \rightarrow V_{\max}$ (substrate-saturated). At $[S] \ll K_M$: $v \rightarrow (V_{\max}/K_M)[S]$ (first-order in substrate).

Lineweaver-Burk plot: $1/v$ vs $1/[S]$ linearizes; extracts $V_{\max} + K_M$.

BST framing: K_M characterizes substrate K-type binding affinity; k_{cat} characterizes K-type transition rate.

Section 6.3 — Glycolysis (Embden-Meyerhof-Parnas)

Glucose $\rightarrow$ 2 pyruvate + 2 ATP + 2 NADH; 10 enzyme steps; near-universal across all life.

Two phases: - Investment (steps 1-5): 2 ATP consumed; glucose $\rightarrow$ 2 G3P - Payoff (steps 6-10): 4 ATP + 2 NADH produced; G3P $\rightarrow$ pyruvate

Net: 2 ATP + 2 NADH per glucose; pyruvate enters TCA in aerobic conditions or fermentation in anaerobic.

BST framing: organized substrate K-type pathway cycle; $10 = N_c^2 + 1 = N_c \cdot n_C - n_C$ BST integer combinations.

Section 6.4 — TCA Cycle (Krebs Cycle, 1937)

Pyruvate (via acetyl-CoA + oxaloacetate $\rightarrow$ citrate) $\rightarrow$ CO₂ + NADH + FADH₂ + GTP; 8 enzyme steps; central metabolic hub.

Per glucose: 2 acetyl-CoA $\times$ (3 NADH + 1 FADH₂ + 1 GTP) = 6 NADH + 2 FADH₂ + 2 GTP per glucose total via TCA.

BST framing: $8 = 2^3$ substrate K-type cycle steps; integer combinatorial pattern.

Section 6.5 — Oxidative Phosphorylation + Chemiosmotic Theory

Electron transport chain (ETC): 4 complexes (I, II, III, IV) pump protons across inner mitochondrial membrane as NADH + FADH₂ donate electrons; O₂ is terminal electron acceptor $\rightarrow$ H₂O.

ATP synthase (complex V): uses proton motive force to phosphorylate ADP $\rightarrow$ ATP; 3 ATP per 10 protons (rotational mechanism, Boyer + Walker Nobel 1997).

Mitchell 1961 chemiosmotic theory (Nobel 1978): proton gradient is universal energy-coupling mechanism. Initially controversial (Slater substrate-level phosphorylation alternative); subsequently established as correct.

BST framing: proton motive force inherits $m_p/m_e = 6\pi^5$ substrate-mechanism (Vol 13 Ch 4); chemiosmotic mechanism = substrate Zone-3 emission of stored gradient energy.

Section 6.6 — ATP Cycle (Energy Currency)

$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{Pi}$: $\Delta G^\circ \approx -30 \text{ kJ/mol}$ (cellular $\Delta G \approx -50 \text{ kJ/mol}$ under typical conditions).

Couples to non-spontaneous biosynthesis (proteins + nucleic acids + lipids), active transport (Na^+/K^+ ATPase, Ca^{2+} ATPase), mechanical work (myosin + actin, kinesin + microtubules).

$\sim 10^8$ ATP molecules per cell per second turnover; human body produces own weight in ATP per day.

BST framing: ATP = substrate Zone-2/Zone-3 energy currency carrier; couples Zone-2 commitment to Zone-3 emission across multiple reaction networks.

Section 6.7 — Connection

- Vol 12 Ch 2-6 Chemistry (foundation chemistry)
- Vol 13 Ch 1-5 Biology (biological context)
- Vol 13 Ch 9 Metabolism (full metabolic treatment)
- Vol 13 Ch 7 Protein Structure (enzyme structural foundation)
- T2417 4-Zone Commitment Cycle (substrate framework)
- Vol 13 Ch 4 DNA Proton Siblings ($m_p/m_e = 6\pi^5$ chemiosmotic cross-link)

Honest scope: Biochemistry-as-Vol-12-chemistry-at-biological-scale is framework-level claim at I-tier; specific substrate-Hamiltonian derivations of enzyme rate enhancements remain multi-year SP-31 program. Standard biochemistry framework (enzymes + metabolism + chemiosmosis + ATP) is well-established mainstream biology with substrate-cartography reading added.

— Lyra, Vol 13 Ch 6 v0.4 chapter-grade narrative REFILLED per Cal #104 + Calibration #23, Saturday 2026-05-23 EDT